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7th ATLAS Machine Learning Workshop

Accelerating the search for mass bumps using BumpNet

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Motivation

- Unfortunately, there is **no new physics** discovered throughout the whole Run 2 period

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- Unfortunately, there is **no new physics** discovered throughout the whole Run 2 period
- Have we explored **everything** from Data?

Existing searches for two-body resonances^[1]

	e	μ	τ	q/g	b	t	γ	Z/W	H	BSM $\rightarrow$ SM ₁ $\times$ SM ₁				BSM $\rightarrow$ SM ₁ $\times$ SM ₂			BSM $\rightarrow$ complex			
										q/g	γ/π^0 's	b	...	tZ/H	bH	...	$\tau qq'$	eqq'	$\mu qq'$	...
e	[37, 38]	[39, 40]	[39]	∅	∅	∅	[41]	[42]	∅	∅	∅	∅		∅	∅	∅	∅	[43, 44]	∅	
μ		[37, 38]	[39]	∅	∅	∅	[41]	[42]	∅	∅	∅	∅		∅	∅	∅	∅	∅	[43, 44]	
τ			[45, 46]	∅	[47]	∅	∅	∅	∅	∅	∅	∅		∅	∅	∅	[48, 49]	∅	∅	
q/g				[29, 30, 50, 51]	[52]	∅	[53, 54]	[55]	∅	∅	∅	∅		∅	∅	∅	∅	∅	∅	
b					[29, 52, 56]	[57]	[54]	[58]	[59]	∅	∅	∅		[60]	∅	∅	∅	∅	∅	
t						[61]	∅	[62]	[63]	∅	∅	∅		[64]	[60]	∅	∅	∅	∅	
γ							[65, 66]	[67–69]	[68, 70]	∅	∅	∅		∅	∅	∅	∅	∅	∅	
Z/W								[71]	[71]	∅	∅	∅		∅	∅	∅	∅	∅	∅	
H									[72, 73]	[74]	∅	∅		∅	∅	∅	∅	∅	∅	
BSM $\rightarrow$ SM ₁ $\times$ SM ₁	q/g									∅	∅	∅		∅	∅	∅	∅	∅	∅	
	γ/π^0 's										[75]	∅		∅	∅	∅	∅	∅	∅	
	b											[76, 77]		∅	∅	∅	∅	∅	∅	
	⋮																			
⋮																				

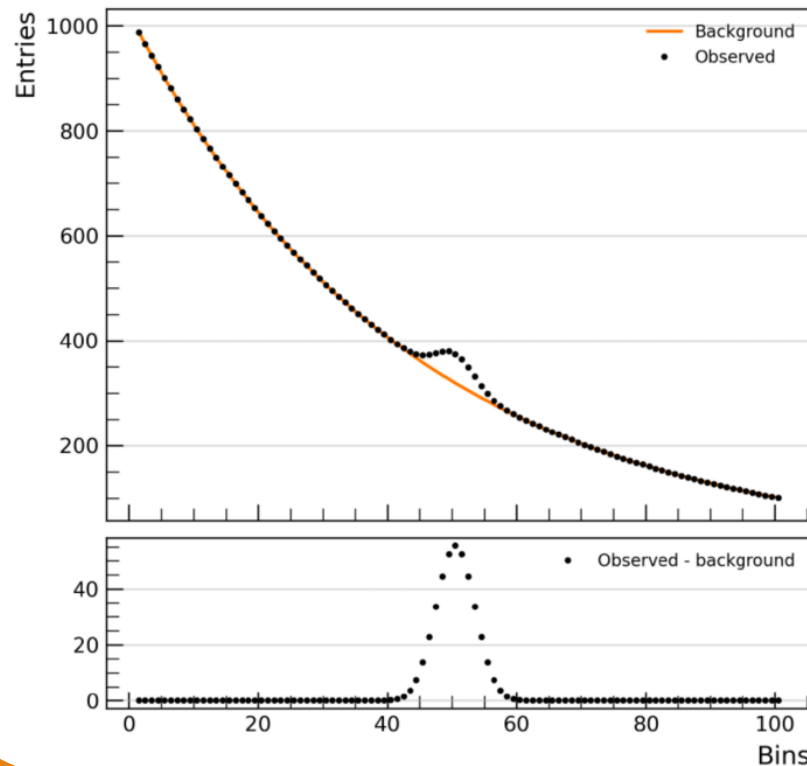
- Why not?
 - **Personpower** and **extremely long** publication process

BumpNet

- A proposal to **scrape systematically** the data in search of **resonant new physics**

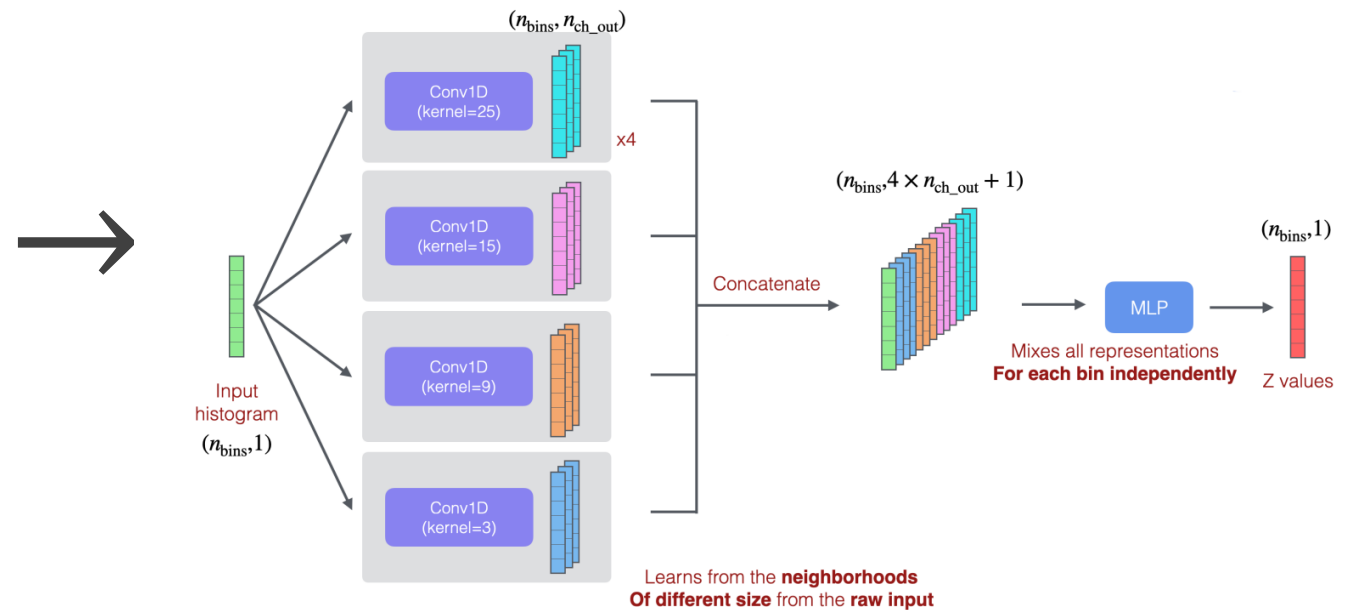
Assumption:

SM background is smoothly falling



NN as the tool:

To search for the deviation as a **bump** due to new physics



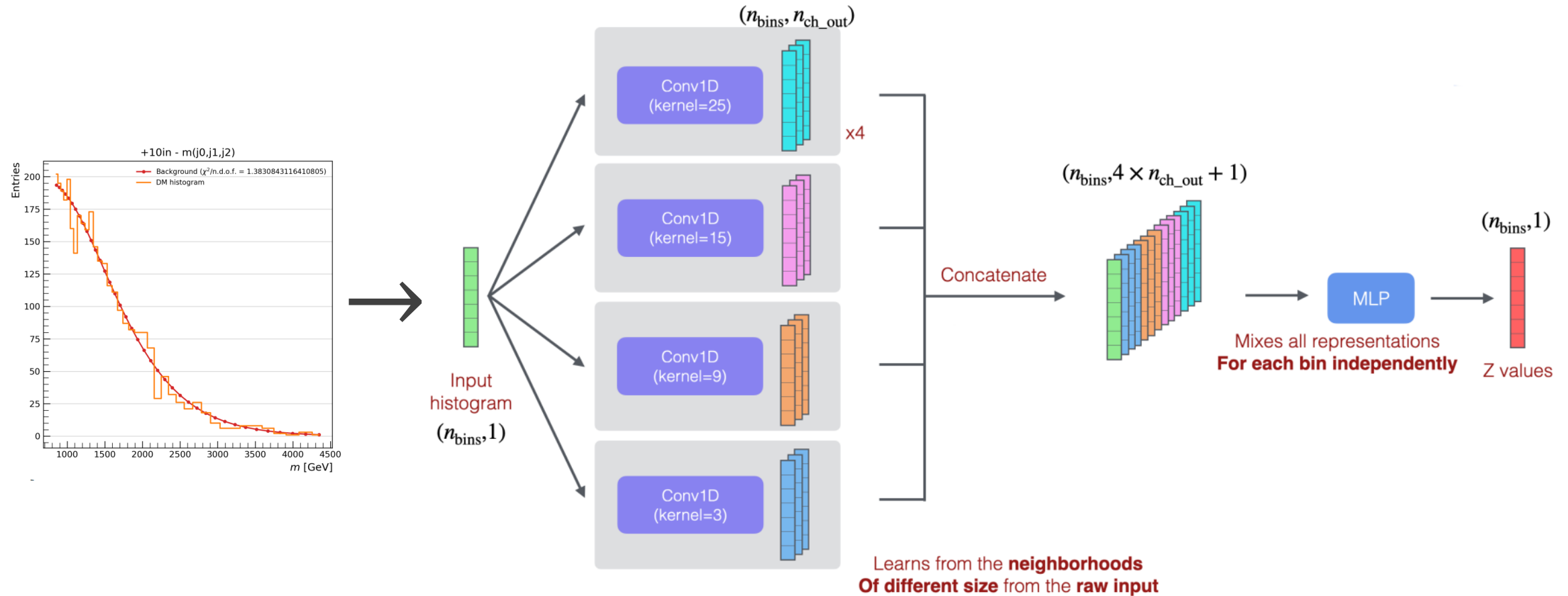
Original idea of BumpNet (DDP) published in arXiv: [2107.11573](https://arxiv.org/abs/2107.11573)

BumpNet - FAQ

- What is the model input?
 - **Mass histograms**: Vector of bin entries of the histograms
- What is the model output?
 - The vector of **statistical significance Z** to find a signal in each mass bin from profile likelihood-ratio test
 - The maximum significance $Z^{\max}$ from each histogram
- What does the model provide?
 - To **identify** the mass spectra (histograms) that give **bump significance** above a certain threshold (e.g., **5-sigma**)
 - Provide **recommendation** on **object selections** and **kinematic cuts** that can be investigate by **complete physics analysis**
- What does the model **NOT** provide?
 - It does **not** perform event-by-event anomaly detection

Neural Network Architecture

- Use of 1D convolution layers, concatenated and passed to a fully connected network
- Intuitive to the number of bins in the histogram



Training dataset

Aim: Teach the network how a realistic **smoothly falling background** looks like

Training sample:

- Histograms **with injected signals** on top of the **background**
- Various **signal strength** and **location**

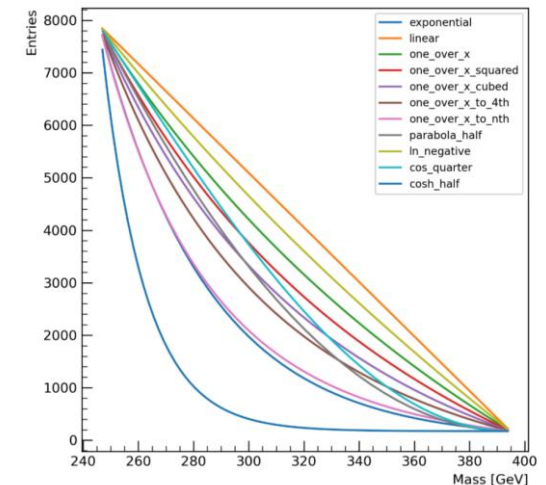
➤ Background:

1. **Analytic functions** (plus stretching) with smoothly descending spectrum
2. **MC simulation** with the same selections as those applied to data
 - **Dark Machines (DM)** dataset
 - **ATLAS** (**ongoing**)

➤ Signal:

1. **Synthetic Gaussian** signals
2. **BSM** samples

$$be^{-ax}, \quad ax + b, \quad \frac{1}{ax} + b, \quad \frac{1}{ax^2} + b, \quad \frac{1}{ax^3} + b, \\ \frac{1}{ax^4} + b, \quad a(x - x_2)^2 + y_2, \quad -a \cdot \ln(x) + b, \\ (y_1 - y_2) \cos(a(x - b)) + y_2, \quad \cosh(a(x - x_2)) + b.$$



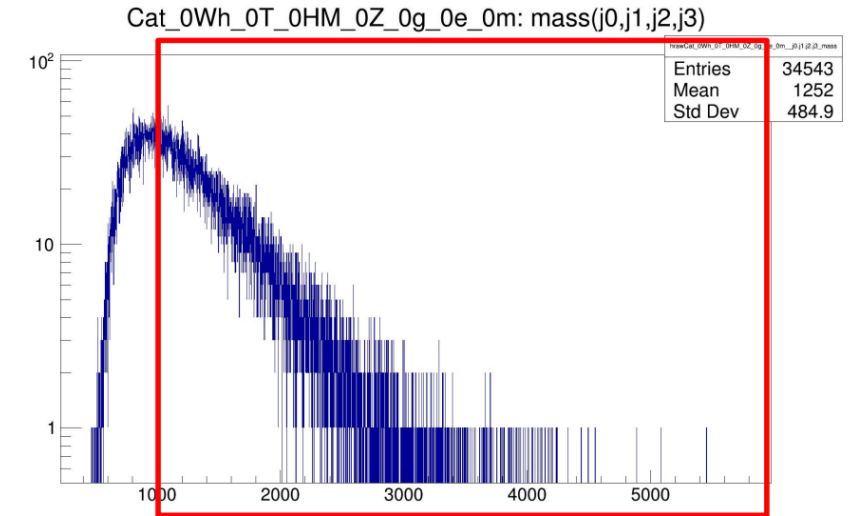
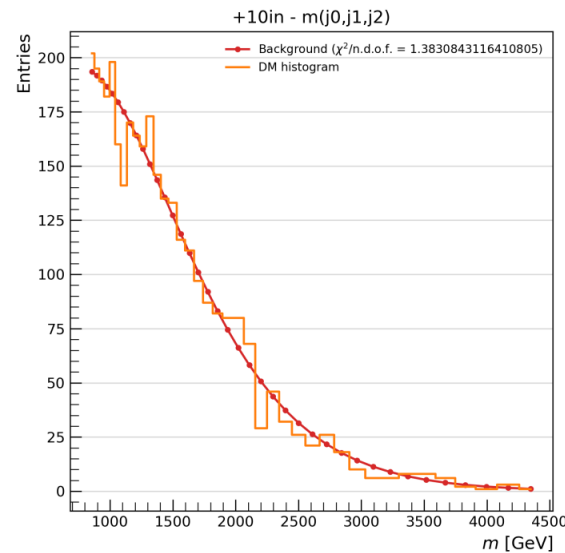
Background: Dark Machines dataset [2]

- Composed of all highest cross-sections **processes at LHC**
- Equivalent to 10 fb^{-1} of 13 TeV data
- Events divided into **channels**
 - **Channel 3:** $H_T \geq 600 \text{ GeV}$ and $\text{MET} > 100 \text{ GeV}$
 - **Channel 2b:** 2ℓ , $H_T \geq 50 \text{ GeV}$ and $\text{MET} > 50 \text{ GeV}$
- Split through all possible **category** and **object combinations**
- Segment further data using **kinematic cuts**, e.g.
 - $\text{MET} > 200$ or 500 GeV
 - Leading object $p_T > 100$ or 200 GeV
- By requiring at least 1500 events per category, total number of histograms:
 - Channel 3: **21k**
 - Channel 2b: **2.8k**

SM processes			
Physics process	Process ID	σ (pb)	$N_{\text{tot}} (N_{10\text{fb}^{-1}})$
$pp \rightarrow jj(+2j)$	njets	$19718_{H_T > 600\text{GeV}}$	415331302 (197179140)
$pp \rightarrow l^\pm \nu_l(+2j)$	w_jets	$10537_{H_T > 100\text{GeV}}$	135692164 (105366237)
$pp \rightarrow \gamma j(+2j)$	gam_jets	$7927_{H_T > 100\text{GeV}}$	123709226 (79268824)
$pp \rightarrow l^+ l^- (+2j)$	z_jets	$3753_{H_T > 100\text{GeV}}$	60076409 (37529592)
$pp \rightarrow t\bar{t}(+2j)$	ttbar	541	13590811 (5412187)
$pp \rightarrow t + \text{jets}(+2j)$	single_top	130	7223883 (1297142)
$pp \rightarrow \bar{t} + \text{jets}(+2j)$	single_topbar	112	7179922 (1116396)
$pp \rightarrow W^+ W^- (+2j)$	ww	82.1	17740278 (821354)
$pp \rightarrow W^\pm t(+2j)$	wtop	57.8	5252172 (577541)
$pp \rightarrow W^\pm \bar{t}(+2j)$	wtopbar	57.8	4723206 (577541)
$pp \rightarrow \gamma\gamma(+2j)$	2gam	47.1	17464818 (470656)
$pp \rightarrow W^\pm \gamma(+2j)$	Wgam	45.1	18633683 (450672)
$pp \rightarrow ZW^\pm(+2j)$	zw	31.6	13847321 (315781)
$pp \rightarrow Z\gamma(+2j)$	Zgam	29.9	15909980 (299439)
$pp \rightarrow ZZ(+2j)$	zz	9.91	7118820 (99092)
$pp \rightarrow h(+2j)$	single_higgs	1.94	2596158 (19383)
$pp \rightarrow t\bar{t}\gamma(+2j)$	ttbarGam	1.55	95217 (15471)
$pp \rightarrow t\bar{t}Z$	ttbarZ	0.59	300000 (5874)
$pp \rightarrow t\bar{t}h(+1j)$	ttbarHiggs	0.46	200476 (4568)
$pp \rightarrow \gamma t(+2j)$	atop	0.39	2776166 (3947)
$pp \rightarrow t\bar{t}W^\pm$	ttbarW	0.35	279365 (3495)
$pp \rightarrow \gamma \bar{t}(+2j)$	atopbar	0.27	4770857 (2707)
$pp \rightarrow Zt(+2j)$	ztop	0.26	3213475 (2554)
$pp \rightarrow Z\bar{t}(+2j)$	ztopbar	0.15	2741276 (1524)
$pp \rightarrow t\bar{t}t\bar{t}$	4top	0.0097	399999 (96)
$pp \rightarrow t\bar{t}W^+ W^-$	ttbarWW	0.0085	150000 (85)

Histograms: Processing and Calibration

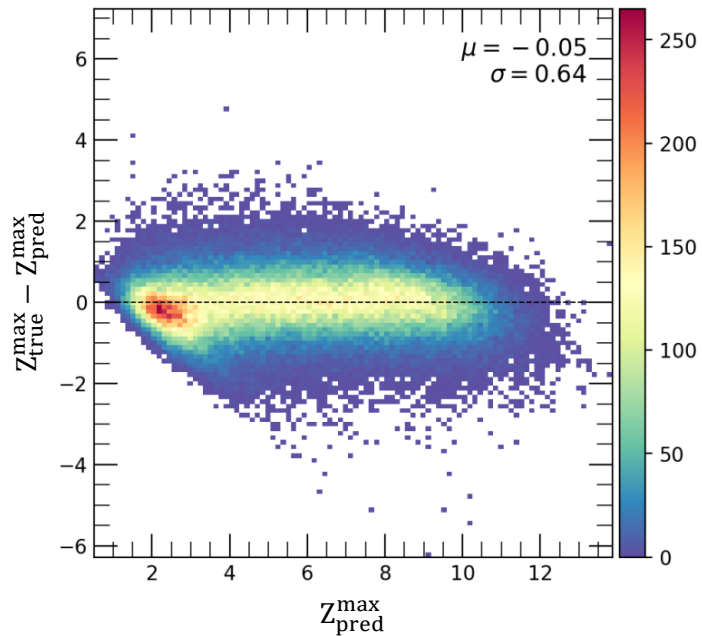
- On MC simulated background
 - So far **done** with **DM dataset**, will need to **fine-tune** for **ATLAS data**
- Zoom in the part **after the maximum** (start smoothly falling)
- Rebinning:
 - **Bin size** should reflect the **detector resolution**
 - Depends on the **objects** involved, their p_T , and η
- **Smooth** the DM histograms



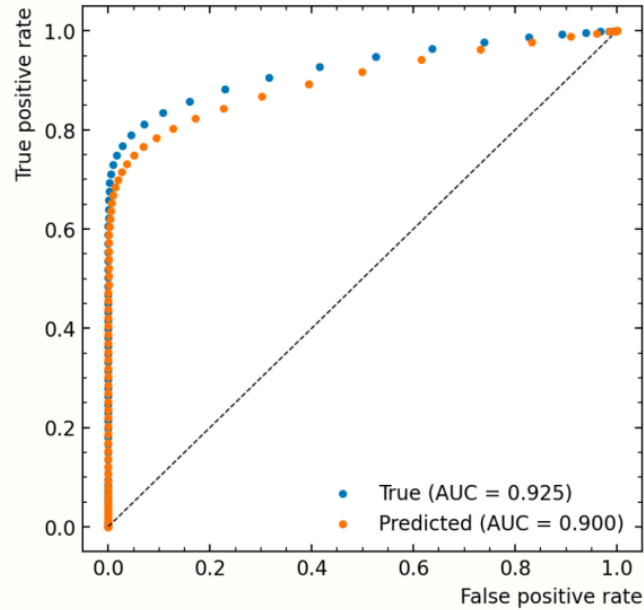
- Add **Poisson fluctuation** to the histogram

Performance Evaluation

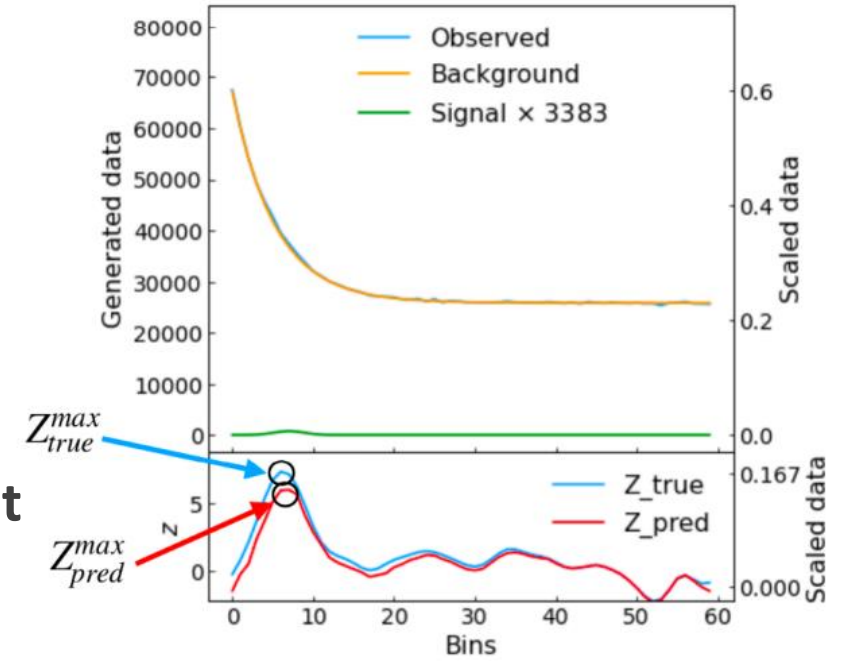
- Tested using unseen data produced in the same manner as training data
- Compute $Z_{\text{true}}^{\text{max}} - Z_{\text{pred}}^{\text{max}}$, where
 - $Z_{\text{true}}^{\text{max}}$: maximum true significance from **Profile Likelihood** test
 - $Z_{\text{pred}}^{\text{max}}$: maximum significance in histogram predicted by **BumpNet**



$Z_{\text{pred}}^{\text{max}}$ **unbiased** with spread of $\approx 0.64\sigma$

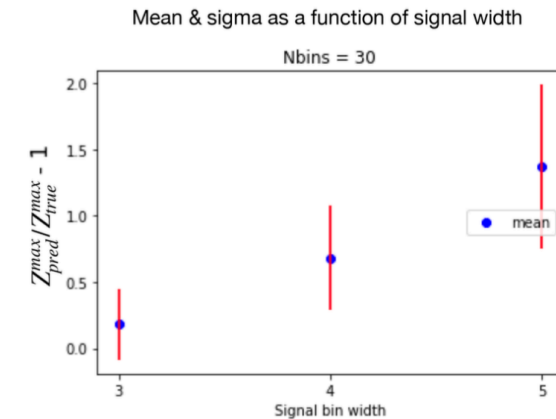
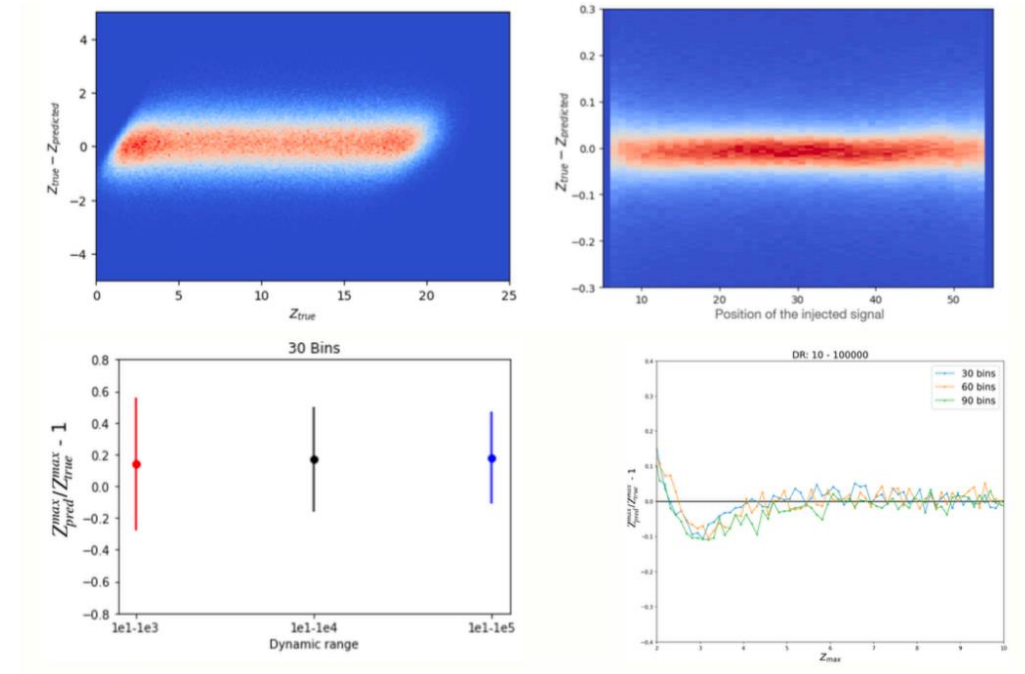


Excellent AUC of 0.9 approaching True(0.925)



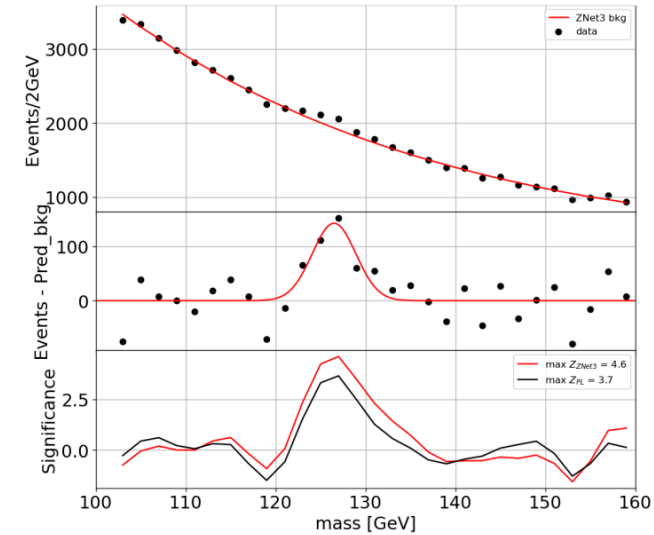
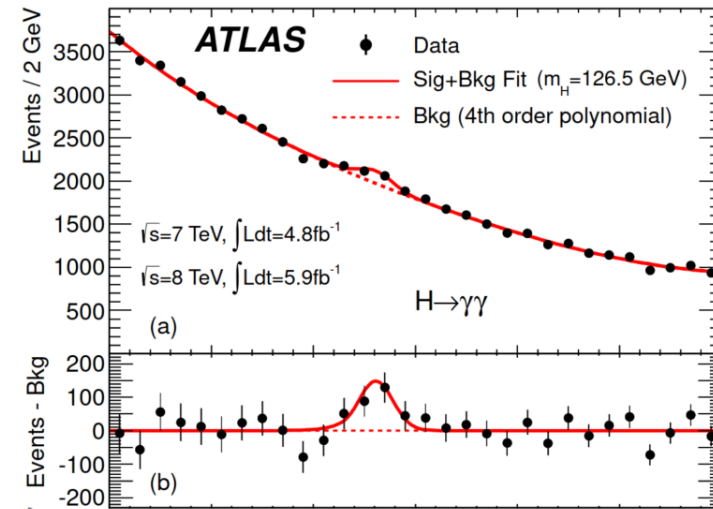
Uncertainties: Performance Stability

- Instead of the experimental uncertainties of data, performance depends more on the **stability of the BumpNet model**
- Tested the model with training samples using different
 - Position of signal
 - Statistics in histograms
 - # of bins
- All result in **stable** performances
- Performance **worsen** for **wider signal resonance**

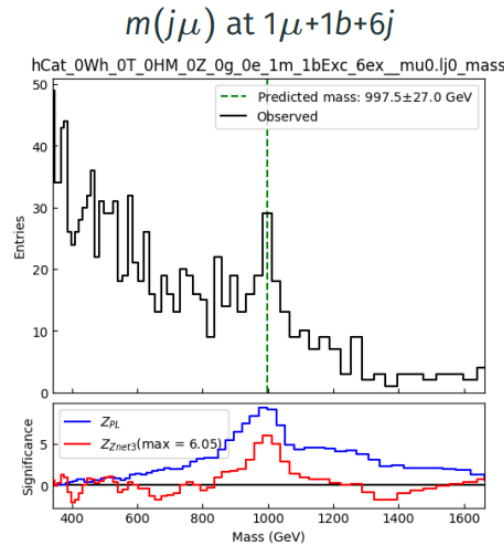


Performance over realistic signals

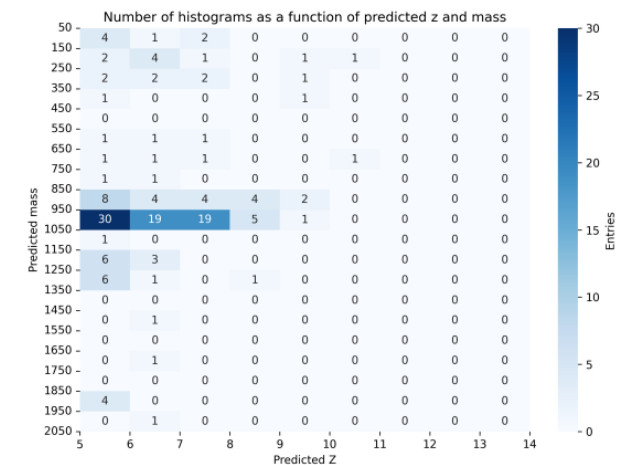
- BumpNet can find the **classic $H \rightarrow \gamma\gamma$ bump** with significance of 4.6σ (true significance from ATLAS is 3.7σ)
- Tested over **simulated BSM signals** added to the Dark Machines background
 - Successfully finds an excess at the expected mass of $\tilde{t}$ at 1TeV
- Other signals tested and successfully found include:
 - $W' \rightarrow WZ$
 - $LQ \rightarrow b e b \mu$
 - $Z' \rightarrow 3\ell$
- Rate of 5σ discoveries in background-only sample at **0.1%**



RPV $\tilde{t} \rightarrow b\ell$

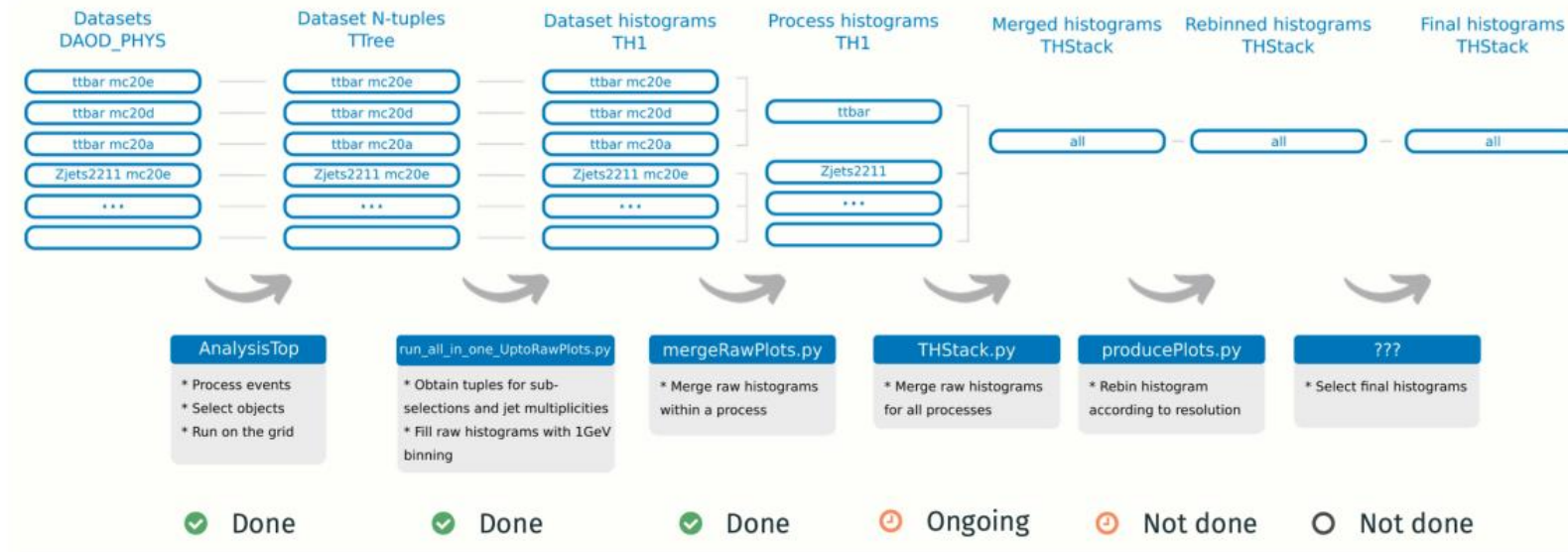


Histograms with $Z_{\text{pred}}^{\text{max}} \geq 5$



Status of ATLAS implementation

- Currently working on workflow to produce mass histograms
 - Processing the **highest cross-section** SM samples in **MC20** with **AnalysisTop**



- Future plans:
 - Move to TopCPToolkit to use **DAOD_PHYSLITE**
 - Produce **BSM signal samples** to test performance
 - **Problem**: Seems most fully simulated BSM samples are only available on **R21**
 - Will test the performance of BumpNet in **real data** in **unblinded final states**

Challenge on ATLAS implementation

- Thought on **systematic uncertainties**

- With DM dataset, BumpNet is proved to be **unsensitive** to systematic uncertainties
- **Large amount** of different background shape in the input can **smear the systematic impact**
- Will apply usual ATLAS systematics to the input background, and **check for the robustness**

- **Interpretation** to the output significance $Z_{\text{pred}}^{\text{max}}$

- Need to precisely estimate the **intrinsic uncertainty** of $Z_{\text{pred}}^{\text{max}}$ to the true significance $Z_{\text{true}}^{\text{max}}$
- Identify uncorrelated histograms with high significance predictions (e.g. bump in same mass for different jet multiplicities)
- Define an **analysis strategy** such that once the NN predicts a 5σ significance at a certain distribution

Summary of BumpNet Analysis

- An **anomaly detection** model
 - Efficient way to **identify resonance bump** in mass histograms
 - For a **complete scan** on **all possible** final states and selections
- The NN (BumpNet) performance is tested with **Dark Machines datasets**
 - Trained with **realistic background** shapes with **injected signals** on top
 - **Excellent performance**: unbiased and small variance from predicted significance to the true significance
 - Working on a (non-ATLAS) **Paper**
- Implementation to **ATLAS datasets**
 - Working on **histogram production** for both background and BSM signal samples
 - Will work on **fine-tuning** the input for the datasets (e.g. binning)
- [Analysis glance page](#)

BackUp

Talk Card

- a) Why this ML idea, rather than traditional methods? What is the state-of-the-art, and is there a theoretical ceiling of performance?
 - Traditional method can't cover large number of final states
 - The state of the art is the direct prediction of the significance
 - The theoretical ceiling is the Profile Likelihood performance
- b) Have you thought about production (e.g. integrating with existing software)? If so, what's the plan?
 - We are currently working on the integration with the ATLAS software scheme to produce invariant mass and massMet distributions.

Talk Card

- c) Have you thought about application to data? How will you handle uncertainties, calibration, etc.?
 - Uncertainty should not be placed on data rather than on the NN prediction.
 - We need to make sure that the max significance predicted is trustable.
 - Reduce the number of outliers and quantify the spread of predicted vs true PL significance.
- d) What are the challenges (i.e. technical, personpower, hardware, conceptual)?
 - The main challenge is to define an analysis strategy such that once the NN predicts, e.g., 5 significance at a certain distribution
 - We will not hesitate to invest the next 3 years of our lives in this exact event selection.